

Oceanic Mixed Bottom Layers

Xing Yu Yi

August 2026

Contents

1	Introduction	2
2	Ekman model	3
2.1	Background	3
2.2	Setting up the model	5
2.3	Numerical implementation	7
2.4	Choice of initial buoyancy profile	8
3	Stokes model	9

Chapter 1

Introduction

In this project, we explore the formation of bottom boundary layer in the abyssal ocean, where water temperature and salinity are nearly uniform, with weak stratification, as opposed to the interior ocean, which has much clearer stratification. These layers range in thickness from 10 to several hundreds of metres.¹ While current theories, such as the “stratified Ekman layer”, can explain the shallower layers,² they cannot account for the massive layers found in the deep, flat ocean plains. These layers are much thicker than the theory predicts, which remains an unsolved problem. Some researchers suggest that breaking internal waves might cause this thickening, but we currently lack a model that can accurately predict the depth based on this idea. In this project, we use simplified theoretical models and computer simulations to explore the physics that determine the depth of these mixed layers.

More specifically, we’ll look at an Ekman model, which looks at the formation of stratified Ekman layers, and a tidal/Stokes model, which looks at the formation of a bottom boundary layer due to a periodic external flow.

¹Armi and Millard Jr [1976](#).

²Weatherly and Martin [1978](#).

Chapter 2

Ekman model

We will begin by looking at an Ekman model for the formation of the bottom boundary layer, where we will have the effects of the Coriolis force with a far-stream velocity $\mathbf{U}_\infty$. The flow will initially be in geostrophic balance, with a pressure gradient to balance this. This model will largely be based off Taylor and Sarkar 2008 and looking at stratified Ekman layers.

2.1 Background

For this project, as we aim to look at the bottom boundary layer of the abyssal ocean, we begin with an introduction to an idealised setup for Ekman layer formation at rigid surfaces.¹ For a stratified fluid, and a constant angular velocity $\mathbf{\Omega}$, we have the following equations of motion by using the *Boussinesq approximation*:

$$\nabla \cdot \mathbf{u} = 0, \quad \frac{D\rho}{Dt} = 0, \quad \frac{D\mathbf{u}}{Dt} + 2\mathbf{\Omega} \times \mathbf{u} = -\frac{1}{\rho_0} (\nabla p + \rho g \mathbf{e}_z) + \mathbf{F}, \quad (2.1)$$

where $\mathbf{u} = (u, v, w)$ is the fluid velocity, ρ is the fluid density, p is the pressure, g is the gravitational field strength, and $\mathbf{F}$ is the external body force. Note that the centrifugal force contribution has been absorbed by the the term involving the body force due to the gravitational field.

In the Boussinesq approximation, we assume that density variations are neglected except in the body force term involving gravity, and the vertical lengthscale of the motion is much smaller than the “scale height” of the medium c^2/g , where c is the speed of sound in the medium. This approximation is good for the oceans, where $c^2/g \sim 200$ km, which is much larger than lengthscale of fluid motion.

¹Kundu, Cohen, and Dowling 2015.

Assuming that the fluid is in hydrostatic balance to simplify the terms in brackets in (2.1),

$$\frac{dp'}{dz} = -g\rho,$$

where we denote perturbed values by primes, e.g. $p = p_0 + p'$ etc, and with the only additional forces acting on the fluid being those due to friction, with kinematic viscosity ν , then the above equations simplify to

$$\frac{Du}{Dt} - fv = -\frac{1}{\rho_0} \frac{\partial p'}{\partial x} + \nu \nabla^2 u, \quad (2.2)$$

$$\frac{Du}{Dt} + fu = -\frac{1}{\rho_0} \frac{\partial p'}{\partial x} + \nu \nabla^2 v, \quad (2.3)$$

$$\frac{Du}{Dt} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial x} - \frac{g\rho'}{\rho_0} + \nu \nabla^2 w, \quad (2.4)$$

which describes fluid motion which is primarily in the horizontal directions on a rotating earth, and locally flat.

We will assume that the flow is initially quasi-steady, in horizontal equilibrium, and so in *geostrophic balance*:

$$-fv = -\frac{1}{\rho_0} \frac{\partial p}{\partial x} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial x}, \quad fu = -\frac{1}{\rho_0} \frac{\partial p}{\partial y} = -\frac{1}{\rho_0} \frac{\partial p'}{\partial y}, \quad (2.5)$$

which is valid for low-Rossby number flows (which is relevant to oceanic flows), where the effects of Coriolis acceleration dominates nonlinear acceleration in this regime, and the *Rossby number* is a nondimensional constant

$$Ro = \frac{\text{Nonlinear acceleration}}{\text{Coriolis acceleration}} \sim \frac{U^2/L}{fU} = \frac{U}{fL}.$$

However, we can note that this assumption breaks down if the Coriolis parameter $f = 2\Omega \sin \theta$ is small (e.g. near the equator, where the latitude, θ , is small) or if friction/unsteadiness becomes important.

Now, consider a steady viscous layer which develops on a flat, rigid, horizontal surface at $z = 0$, and there is a far-stream flow $\mathbf{U} = U\hat{\mathbf{x}}$ far from the surface.

Since the fluid is initially in geostrophic balance, the pressure gradient will balance the Coriolis acceleration

$$fU = -\frac{1}{\rho} \frac{\partial p}{\partial y}.$$

Near the rigid surface, inside the boundary layer, where friction is important, we have

$$-fv = \nu \frac{d^2 u}{dz^2}, \quad \text{and} \quad fu = \nu \frac{d^2 v}{dz^2} + fU. \quad (2.6)$$

We impose kinematic and no-slip boundary conditions at the wall, and matching the far-stream velocity as $z \rightarrow \infty$. Introducing the *complex velocity* $V = u + iv$,

$$\frac{d^2 V}{dz^2} = \frac{if}{\nu}(V - U) \quad \Rightarrow \quad V = U \left(1 - \exp \left\{ -\frac{(1+i)z}{\delta} \right\} \right) \quad (2.7)$$

noting that $V \rightarrow U$ as $z \rightarrow \infty$, and $V = 0$ at $z = 0$. We have also introduced $\delta = \sqrt{2\nu/f}$, which is the *e*-folding length, noting that this is independent of U . This gives us an *Ekman layer*, and we can also plot a *hodograph*, which shows the *Ekman spiral*.

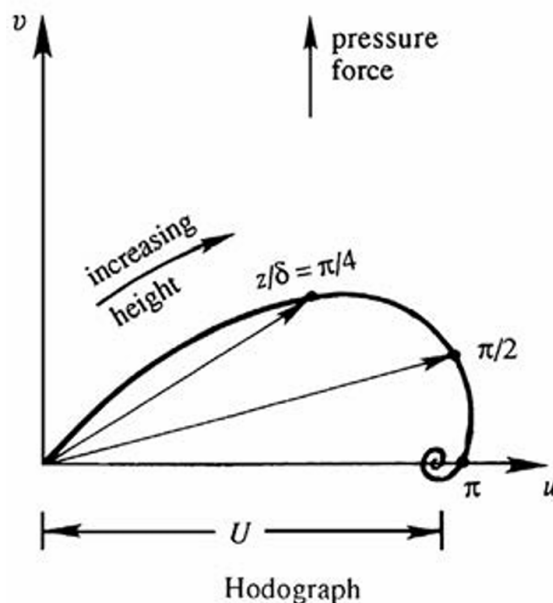


Figure 2.1: Hodograph of the analytic solution for an Ekman layer above a rigid surface with steady far-stream velocity U in the x -direction.

2.2 Setting up the model

For the Ekman model for the formation of the turbulent bottom boundary Ekman layer, we begin with an initially steady flow in geostrophic balance in the domain $z \geq 0$, with a flat, rigid, adiabatic boundary at $z = 0$. In this model, we will adjust the outer layer stratification by adjusting our initial buoyancy profiles, and see how buoyancy evolves over time. We will make some approximations which will simplify our equations:²

²Taylor and Sarkar 2008.

- The free stream is in geostrophic balance and aligned in the x -direction.
- The seafloor is modelled as a flat, rough surface.
- The roughness elements are negligibly small to be resolved by the grid, but with their effects parameterised by a *near-wall model*.
- The lateral boundaries are periodic, which is consistent with the assumption that the flow is statistically homogeneous in the horizontal plane.
- The domain is sufficiently large so that the flow is decorrelated over a distance equal to the domain size.

Taking the free-stream velocity $U_\infty = 0.04 \text{ m s}^{-1}$ and Coriolis parameter $f = 10^{-4} \text{ rad s}^{-1}$, we have outer unit $D = U_\infty/f = 400 \text{ m}$, where our domain has size comparable to $\sim 0.1D$ in the horizontal directions. Due to horizontal periodicity, the mean vertical velocity must be zero. Hence, the Ekman layers formed here will be unaffected by Ekman pumping/suction from large-scale horizontal gradients.

We aim to solve the following set of Navier-Stokes equations:

$$\frac{\partial u}{\partial t} + \mathbf{u} \cdot \nabla u - fv = -\frac{\partial p}{\partial x}, \quad (2.8)$$

$$\frac{\partial v}{\partial t} + \mathbf{u} \cdot \nabla v + fu = -\frac{\partial p}{\partial y} + fU_\infty, \quad (2.9)$$

where U_∞ is the free-stream velocity in the x -direction. The additional forcing term on the RHS in (2.9) is due to the initial geostrophic balance.

At the bottom boundary, we impose a no-flux boundary condition, and ignoring the negligibly small viscous sublayer, we use the quadratic drag boundary condition

$$\frac{\boldsymbol{\tau}_w}{\rho_0} = -c^D |\mathbf{u}| \mathbf{u},$$

where $\boldsymbol{\tau}_w$ is wall stress, and c^D is the coefficient of drag, given by

$$c^D = \left(\frac{\kappa}{\log(z_1/z_0)} \right)^2$$

where z_1 is the distance of the first grid point away from the wall, z_0 is the friction length, and $\kappa \approx 0.41$ is the von Karman constant. In the above, we have used the law of the wall, where, in the small viscous sublayer, we have

$$|\mathbf{u}| = \frac{u_*}{\kappa} \log \left(\frac{z}{z_0} \right),$$

where $u_* = \sqrt{\boldsymbol{\tau}_w/\rho_0}$ is the friction velocity.

Above the free surface, we include an additional sponge layer, to reduce the effects of an internal wave reflection at the surface, thus reducing the effect that this has on the Ekman layer formation. This sponge layer will relax the buoyancy profile to some chosen background initial profile $b_i(z)$, which we will evolve as an active tracer.

2.3 Numerical implementation

We use the Oceananigans package in the Julia programming language to solve for our flows, and initially, the numerical setup is similar to that of Taylor and Sarkar 2008. We checked that our results agreed closely with the case of linear stratification, before proceeding to adjust other parameters and seeing their effects on these stratified Ekman layers.

For our numerical implementation, we will use large eddy simulations (LES) to accurately solve the Navier-Stokes equations, where we aim to accurately solve a low-pass-filtered version of the Navier-Stokes equations. However, since the filtered version of the nonlinear advection terms involve both total and filtered velocity fields, this results in fewer equations than unknowns. This is the well-known turbulence closure problem, so we need to model residual stresses in terms of filtered quantities, in which we will use an anisotropic minimum dissipation method (given by the turbulence closure `AnisotropicMinimumDissipation`), and we also account for the molecular kinematic viscosity and molecular diffusivity for the ocean (using the turbulence closure `ScalarDiffusivity`).³⁴

For our advection scheme, we use the third-order Runge-Kutta method, and we use a stretched grid, with finer grid spacing nearer to the wall to see the mixed layer in greater resolution.

In order to be able to compare the tidal model (see Chapter 3), the following parameters will be fixed:

- $U_\infty = U_0 = 0.04 \text{ m s}^{-1}$
- $\omega = f = 10^{-4} \text{ s}^{-1}$
- $\nu = 10^{-6} \text{ m}^2 \text{ s}^{-1}$
- $Pr = \nu/\kappa = 10 \implies \kappa = 10^{-7} \text{ m}^2 \text{ s}^{-1}$

³Verstappen 2018.

⁴Vreugdenhil and Taylor 2018.

2.4 Choice of initial buoyancy profile

The following profiles have been considered, where we denote zH to be domain height, T to be where some profiles have the sharpest change in buoyancy at $z = T$, and L to be the decay length:

•

$$b_i(z) = N^2 z \left(1 - \frac{1}{1 + z^n / ((n-1)T^n)} \right)$$

- This profile has zero gradient at $z = 0$.
- This decays to a linear profile for large z , albeit somewhat slowly.

•

$$b_i(z) = N^2 z \left[1 - \exp \left(-\frac{z^n}{nT^n} \right) \right]$$

- This profile has zero gradient at $z = 0$, and decays to a linear profile for large z quickly.
- This profile is likely too complicated to analyse.

•

$$b_i(z) = N^2 [z + L (e^{-z/L} - 1)]$$

- This profile reaches a linear profile quickly for large z .
- There is little control over where sharp changes in buoyancy occur.

•

$$b_i(z) = \Delta b \left[\frac{L(e^{z/L} - 1) - z}{L(e^{H/L} - 1) - H} \right]$$

- This profile has fixed buoyancy difference $\Delta b = b(H) - b(0)$ between $z = 0$ and $z = H$.
- As $L \rightarrow \infty$, or at least for large L , this profile approaches $\Delta b (z/H)^2$.

•

$$b_i(z) = \frac{N^2}{S} \log [1 + e^{S(z-T)}]$$

- This profile smoothens out a piecewise linear profile which changes sharply at $z = T$ to a linear profile with gradient N^2 .
- This profile is easier to control and adjust.

Chapter 3

Stokes model

We will next look at the Stokes model

Bibliography

- [1] Laurence Armi and Robert C Millard Jr. “The bottom boundary layer of the deep ocean”. In: *Journal of Geophysical Research* 81.27 (1976), pp. 4983–4990.
- [2] Georges L Weatherly and Paul J Martin. “On the structure and dynamics of the oceanic bottom boundary layer”. In: *Journal of Physical Oceanography* 8.4 (1978), pp. 557–570.
- [3] John R Taylor and Sutanu Sarkar. “Stratification effects in a bottom Ekman layer”. In: *Journal of Physical Oceanography* 38.11 (2008), pp. 2535–2555.
- [4] Pijush K Kundu, Ira M Cohen, and David R Dowling. *Fluid Mechanics—Sixth Edition*. 6th ed. Elsevier, 2015. ISBN: 9780124059351.
- [5] Roel Verstappen. “How much eddy dissipation is needed to counterbalance the nonlinear production of small, unresolved scales in a large-eddy simulation of turbulence?” In: *Computers & Fluids* 176 (2018), pp. 276–284.
- [6] Catherine A Vreugdenhil and John R Taylor. “Large-eddy simulations of stratified plane Couette flow using the anisotropic minimum-dissipation model”. In: *Physics of Fluids* 30.8 (2018).